LABORATORY REPORT: DIVISION OF FORENSIC BIOLOGY

Case Reference: W-2026-09
Date of Analysis: October 7, 2026
Examiner: Ummay Habiba, Forensic Biology Specialist (In Silico)

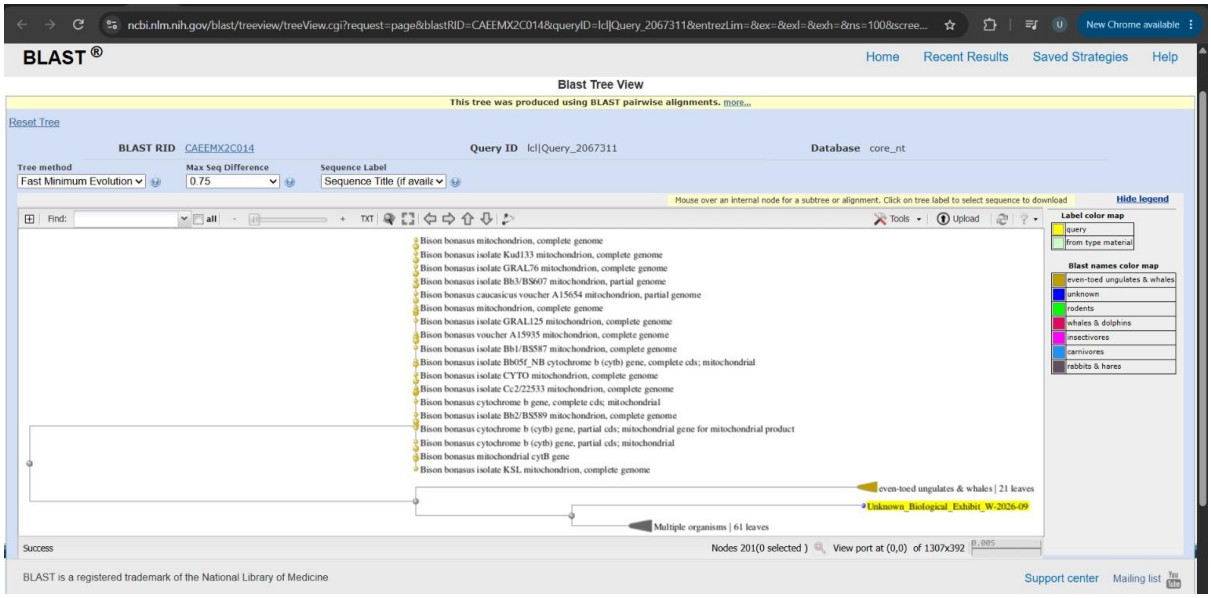
1. OBJECTIVE OF EXAMINATION
To perform genetic sequence alignment and molecular taxonomic classification on an unlabelled biological matrix (Specimen #W-2026-09) seized during an anti-poaching and illegal meat trade investigation.

2. MATERIALS AND METHODOLOGY

- **Database Utility:** National Center for Biotechnology Information (NCBI) BLAST framework.
- **Algorithm Parameters:** Fast Minimum Evolution tree construction via pairwise alignment calculations.
- **Target DNA Fragment:** Purified mitochondrial sequence segment (Cytochrome b / partial genome locus).

3. OBSERVATIONS & DATA ANALYSIS
Upon conducting a structural tree alignment, the unknown sample clustered directly into the Cetartiodactyla clade.

- **Identified Clade:** Even-toed ungulates & whales [1]
- **Immediate Node Grouping:** *Bison bonasus* (European Bison) [1]
- **Phylogenetic Clustering:** The query sequence sits on a terminal branch directly linked to established reference sequences of *Bison bonasus* with a maximum sequence difference threshold of 0.75.



4. FORENSIC CONCLUSION

Based on the absolute taxonomic clustering illustrated in the generated BLAST Pairwise Alignment Tree, the unknown biological evidence sample (Specimen #W-2026-09) is positively identified as originating from a **European Bison (*Bison bonasus*)**. This visual cluster satisfies the scientific burden of proof required to establish species identity for legal proceedings regarding illegal wildlife trafficking.

Signature of Examiner:
[Ummay Habiba]
